Local versus global learning in a data-constrained mouse V1 circuit: population geometry and its match to Neuropixels recordings

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Abstract

Does the way synapses learn leave a measurable signature in what cortical populations represent? We built a recurrent model of a mouse V1 microcircuit from 1,000 differentiable firing-rate models fitted to patch-clamp recordings, wired with measured cell-type connection probabilities and driven by the same visual stimuli as Allen Neuropixels recordings. Identical circuits were trained either with a cell-local, forward-in-time rule (e-prop, with or without the cell's spike-history term) or with exact back-propagation through time, under the same optimizer, on two objectives: matching in-vivo response statistics, and a change-detection task. Their responses to 118 natural scenes never used in training were compared with 627 mouse V1 units, under a protocol locked before the held-out cohort of fitted cells was used (10 paired networks per condition).

Reliability-corrected representational similarity with mouse V1 ($\approx$0.38) was the same for every learning rule and for untrained networks, under both objectives. When matching in-vivo statistics, the local rule fitted them more closely than BPTT (loss 0.31 vs 1.04), but by scaling recurrent weights about 90-fold. This collapsed natural-image dimensionality (participation ratio 12 vs 31; mouse 32) and decodability (20% vs 79%; mouse 84%), consistently across all networks. At matched loss, however, the two rules produced the same geometry. A behavioral objective changed natural-scene geometry little under any rule.

In this circuit, the population geometry shared with mouse V1 is fixed by its biological constraints. The learning rule matters mainly through which solutions it can reach, so learning rules should be compared at matched performance and with untrained baselines.

1 Introduction

Cortical synapses must change in ways that improve behaviour, using only information available locally. Learning rules that respect this constraint, such as three-factor rules with eligibility traces and broadcast error signals (Gerstner et al., 2018; Murray, 2019; Bellec et al., 2020), can solve tasks that otherwise need back-propagation through time (BPTT). They differ from exact gradients in how credit is assigned (Lillicrap et al., 2020; Richards et al., 2019). If brains learn with such rules, the difference should leave traces in what a circuit ends up representing. Representations of networks trained with different rules can indeed be told apart from their observables (Nayebi et al., 2020). Whether these signatures appear in biologically constrained circuits, and whether they move a model towards or away from real cortex, is unknown.

Mouse primary visual cortex (V1) offers the data needed to ask this. There are fitted single-neuron models for hundreds of recorded cells (Yeung et al., 2025), cell-type-specific connectivity from electron microscopy and synaptic physiology (Campagnola et al., 2022), and large-scale Neuropixels recordings of responses to the same natural images and gratings (Siegle et al., 2021; de Vries et al., 2020). Building on these, Ito et al. (2026) trained a large V1 model with BPTT to match in-vivo activity statistics. That work leaves open the question we address: does the learning algorithm itself shape the population geometry, and is a cell-local rule closer to or further from mouse V1 than the exact gradient?

Our earlier version of this question used abstract working-memory tasks. Its conclusions were dominated by the task and by the cells' slow intrinsic filters, and its inputs had no counterpart

in any recording. Here we rebuilt the setup around the data. A V1 microcircuit of 1,000 patch-clamp-fitted cells with measured connectivity receives the same visual stimuli as the recorded mice, through a fixed LGN front end. The same circuit is trained with exact BPTT or with a cell-local e-prop rule, with or without the cell's own spike-history term. Both use the same optimizer, learning rates and budget, under two different objectives: matching in-vivo response statistics, and a Visual-Behavior-style change-detection task. We then compare population responses to 118 natural scenes that were never used in training with those of mouse V1 units. We use representational similarity, CKA, dimensionality, sparseness and decoding, correct for the reliability of both model and data, and relate geometry to task performance along training. The protocol was locked before the held-out cohort of fitted cells was used.

The main results are:

- The reliability-corrected similarity to mouse V1 was the same for every rule and for untrained networks. The shared geometry is set by the circuit's constraints, not by its learning rule.
- With in-vivo statistics as the objective, the local rule fitted them more closely than BPTT, but only by moving into a regime with very large weights. There natural-image dimensionality, decodability and sparseness departed from the mouse. At matched loss, the two rules' geometry was the same.
- A behavioral objective changed the natural-scene geometry little under any rule.

2 Results

2.1 A data-constrained V1 circuit and the mouse reference

Each network had 1,000 fitted GFR cells wired with measured cell-type connection probabilities. It was driven by an LGN front end and trained with a cell-local e-prop rule (with or without the spike-history term β) or with exact BPTT, with the same optimizer and learning rates. The mouse reference was 627 V1 units from six Neuropixels sessions in six mice. Their population geometry for the 118 natural scenes is highly reliable: the split-half reliability of the RDM was 0.95, two disjoint halves of the units gave RDMs correlated at $\rho = 0.80$, the participation ratio (200 units) was 32, and single-repeat image decoding reached 84%. All model results below come from the confirmation cohort, with 10 paired networks per condition, analysed as locked.

2.2 Training reshapes tuning statistics

Training on the data objective brought the networks' grating responses towards the in-vivo distributions (Fig. 1). Mean tuning-distribution similarity rose from 0.41 for untrained networks to 0.57 (BPTT) and 0.59 (local); the two rules did not differ significantly. Median preferred-direction rates of Pyr cells fell from 5.4 Hz to 0.9 Hz under both rules (in vivo 1.45 Hz), and their median OSI rose from 0.13 to 0.32 (BPTT) and 0.21 (local), against 0.29 in vivo. The local rule reached a much lower final data loss than BPTT: 0.31 vs 1.04, from 5.68 untrained. BPTT became unstable late in training at the shared learning rate.

2.3 Reliability-corrected RSA does not depend on the learning rule

The preregistered primary endpoint is Spearman RSA between model and mouse natural-scene RDMs, corrected for the reliability of both. It was essentially the same for every condition: 0.38 for untrained networks, and between 0.36 and 0.39 for every trained network under both objectives (Fig. 2a,g). No contrast approached significance; the largest was local vs no β on data matching, -0.028, $p = 0.10$. Local vs BPTT differed by -0.005 [-0.033, 0.026] (data matching) and $+0.003$ [-0.001, 0.008] (change detection). Together with the untrained value, this shows that the image-similarity structure the models share with mouse V1 is set by the fixed front end and the fitted cells, not by how the recurrent synapses were trained.

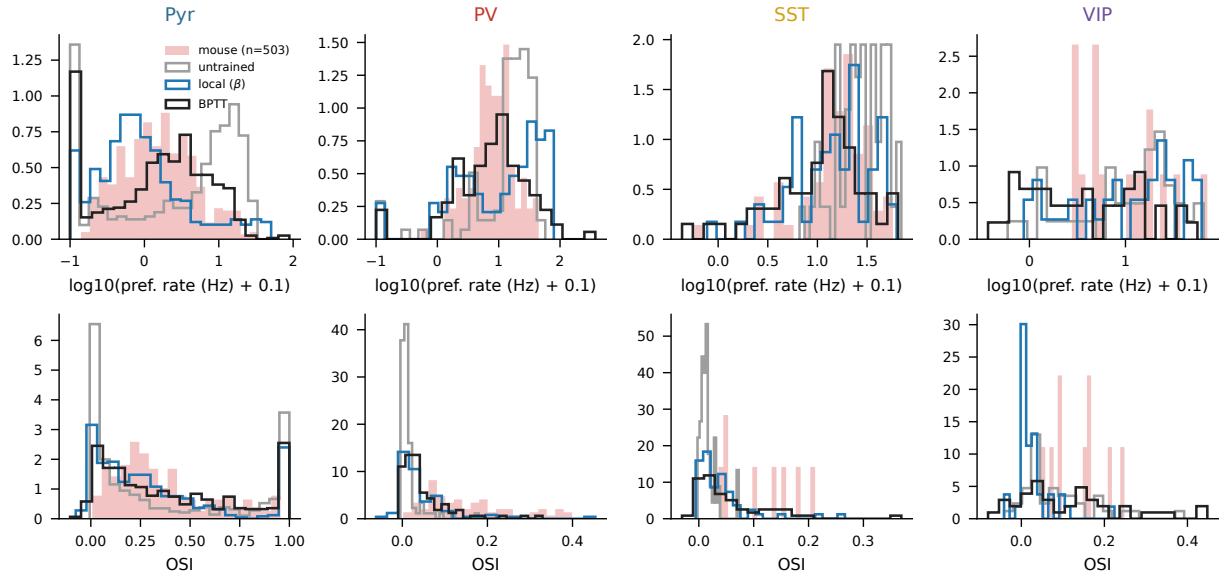


Figure 1: Tuning distributions after data-matching training (one confirmation network). Shaded: mouse V1 Neuropixels units; lines: model cells of the same type before training and after local or BPTT training. Top: preferred-direction rate; bottom: OSI of responsive cells.

2.4 Data matching: the local rule overshoots into an image-degraded regime

Under data matching, the secondary measures separated the rules sharply, in the same direction in all 10 paired networks (Fig. 2b–e). Each of these contrasts has unadjusted $p = 0.002$, the minimum for 10 pairs. After Holm correction over the full secondary family (∼40 tests) the adjusted value is $p = 0.078$.

Compared with BPTT, local-trained networks had:

- raw RSA with the mouse 0.13 vs 0.32;
- CKA 0.26 vs 0.52;
- lower dimensionality: participation ratio 12 vs 31, against 32 in the mouse;
- much poorer image decoding: 20% vs 79%, against 84% in the mouse;
- less mouse-like lifetime sparseness: similarity 0.57 vs 0.75.

Their natural-scene RDMs were also unreliable across repeats (0.14 vs 0.77). This is why the reliability-corrected RSA was unchanged while the raw RSA fell.

The cause is visible in the circuit. On average the local rule multiplied the recurrent magnitudes by a relative change of 87, against 2.5 for BPTT. It also drove PV cells to 43 Hz during natural scenes (BPTT 8 Hz). Removing β from the eligibility trace made no detectable difference to any geometry measure.

At matched performance the rules agree. Representations along training were related to the data loss at each checkpoint (Fig. 3a,b). At equal loss the local and BPTT trajectories coincide. At the loss BPTT reached at its best (≈1.5), local networks (step 200, loss 1.39) had RSA 0.34 and CKA 0.54, the same as BPTT (0.31–0.34 and 0.51–0.54). The local networks' geometry collapsed only after they continued past that point, fitting the grating and gray statistics more closely than BPTT ever did. The divergence is therefore a property of the solutions the local rule can reach, not a signature of the rule at matched performance.

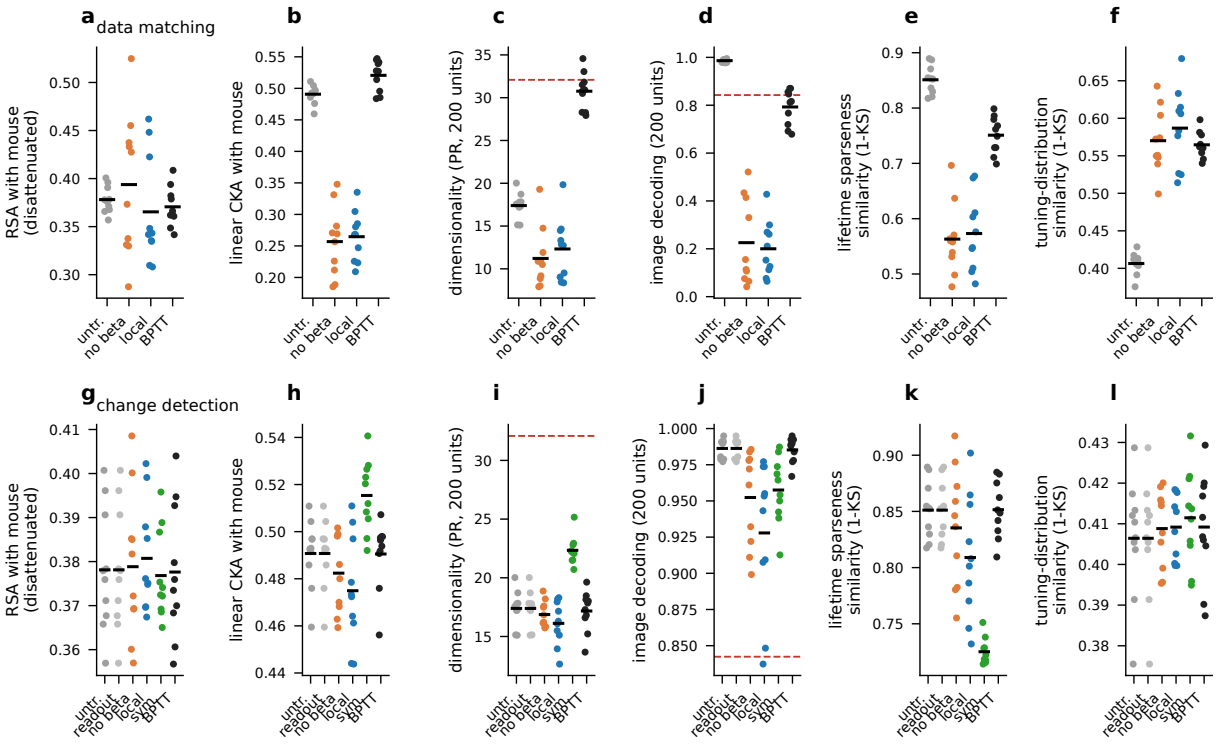


Figure 2: Population geometry for 118 held-out natural scenes, compared with mouse V1. Each dot is one network (10 per condition); bars show means; red dashed lines show the mouse value. (a,g) Reliability-corrected RSA (primary endpoint). (b,h) Linear CKA. (c,i) Participation ratio, 200 units. (d,j) Single-repeat image decoding, 200 units. (e,k) Lifetime-sparseness similarity. (f,l) Mean tuning-distribution similarity. Top: data matching; bottom: change detection.

2.5 Change detection: small, rule-specific changes on top of a fixed geometry

In change detection, the local rule reached higher validation accuracy than BPTT: 0.76 vs 0.68 (readout only 0.62). Symmetric-feedback local reached 0.84. Natural-scene geometry changed little under every rule: the correlation of each network's RDM with its own untrained RDM was 0.99 (BPTT), 0.92 (local) and 0.94 (symmetric feedback). Two consistent effects appeared, both in 10/10 networks with unadjusted $p = 0.002$ (Holm-adjusted $p = 0.125$):

- local learning slightly lowered image decoding (93% vs 98.5% for BPTT);
- local learning with symmetric feedback raised dimensionality (PR 22 vs 16) and CKA with mouse V1 (0.52 vs 0.47).

3 Discussion

We asked whether training a biologically constrained V1 circuit with a cell-local rule rather than the exact gradient leaves a measurable trace in its population geometry, and which brings it closer to mouse V1. Four conclusions follow from the confirmatory data.

The shared image geometry is set before learning. The reliability-corrected agreement with mouse V1 on 118 natural scenes was the same for untrained networks and for every learning rule, under both objectives. In this circuit, what natural images look like to the population is determined by the front end, the fitted cells and the measured wiring. That is consistent with models in which V1 selectivity is largely feedforward (Lien and Scanziani, 2018). It also cautions against reading RSA agreement as evidence for any particular learning rule. An untrained,

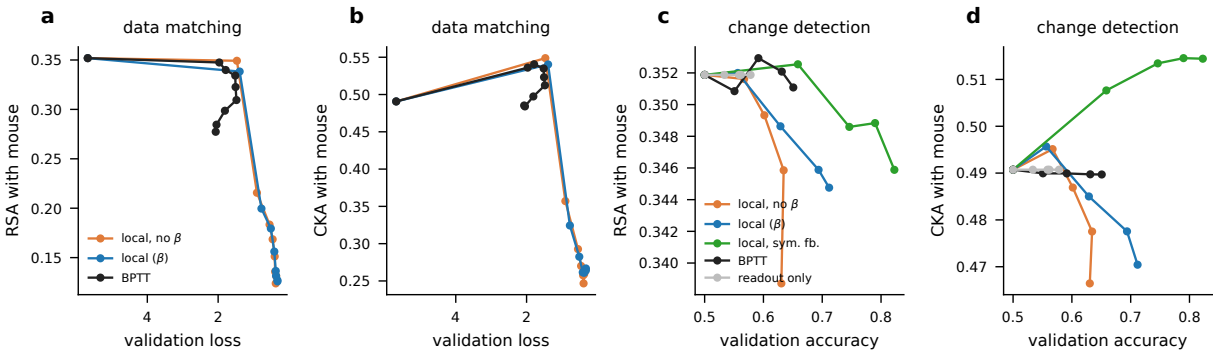


Figure 3: Geometry along training. Mean RSA (raw) and CKA with mouse V1 at each saved checkpoint, plotted against that checkpoint's validation value (10 networks per rule). (a,b) Data matching (loss decreases to the right). (c,d) Change detection (accuracy).

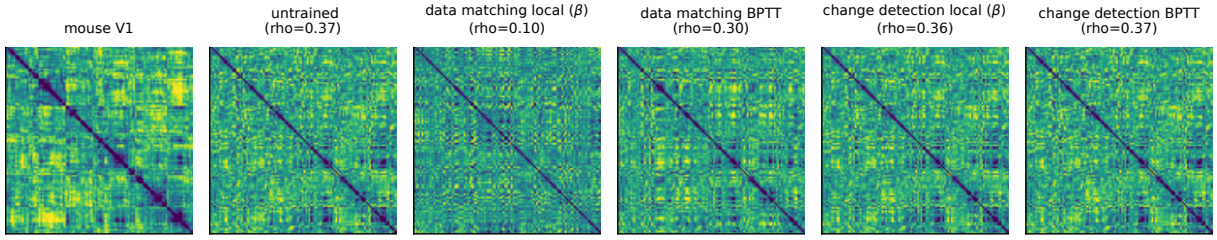


Figure 4: Natural-scene RDMs (one confirmation seed). Images are ordered by hierarchical clustering of the mouse responses. Titles give the Spearman correlation with the mouse RDM.

well-constrained circuit already achieves it, so an untrained baseline and reliability correction are essential.

Rules differ in where they can take the circuit. With matching in-vivo statistics as the goal, the local rule optimized the trained statistics further than BPTT. It did so by multiplying recurrent weights roughly 90-fold and driving interneurons hard. This solution generalized poorly to natural images: dimensionality, decodability and sparseness moved away from mouse values. BPTT stopped earlier and retained mouse-like dimensionality and decodability. Crucially, at matched loss the two rules produced the same geometry. The difference is therefore not an intrinsic signature of local credit assignment but a consequence of which regions of weight space each rule can reach. Local rules ignore network feedback in their gradient estimates and can keep pushing in a direction that the exact gradient would correct. This argues for adding mechanisms that bound such drift, such as weight or rate homeostasis, when local rules are used to fit data, and for comparing learning rules at matched performance.

On a behavioral task the representation barely moves. Training change detection on eight images changed natural-scene geometry little under any rule. The local rule solved the task better than BPTT at the shared learning rates, and with symmetric feedback it raised dimensionality towards the mouse value. These effects are small compared with the mouse–model gap.

The spike-history term. Unlike in our earlier abstract tasks, removing β from the eligibility trace did not change any geometry measure here. Its importance appears to depend on whether the task requires credit assignment through the cells' adaptation dynamics.

Relation to the specification's hypotheses. We did not find that local learning yields representations "significantly more" similar to mouse V1 than BPTT. On the preregistered endpoint

the rules were indistinguishable, and on several secondary measures BPTT was closer to the mouse under data matching.

Limitations.

- *Front end and fixed inputs.* The LGN model and the LGN→V1 weights are simplified and fixed. Plastic thalamocortical weights would give learning more leverage over geometry.
- *Rates, not spikes; 20-ms bins.* Fano factors and spike-timing statistics cannot be compared with the recordings.
- *Scale and cells.* The circuit has 1,000 cells sampled with replacement from about 300 fits, and represents one L2/3-like column without laminar structure.
- *Learning rates.* Rates were shared across rules and chosen on a small development grid. BPTT was unstable at the chosen data-matching rate late in training. The change-detection recurrent rate was not tuned. Matched-performance comparisons address but do not remove this dependence.
- *Mouse data.* We used six sessions (627 V1 units) recorded during passive viewing. Units were pooled across animals into a pseudo-population, and putative cell classes were assigned by waveform only.
- *Multiplicity.* The secondary contrasts were consistent across all 10 networks but do not survive Holm correction over the full secondary family.
- *Disattenuation.* When a model's RDM reliability is low (0.14 for local data-matching networks), the corrected RSA is noisy.

4 Methods

4.1 Circuit

Each network contains 1,000 cells: 800 pyramidal (Pyr), 100 PV, 60 SST and 40 VIP. The cells are differentiable generalized firing-rate (GFR) models fitted to Allen Cell Types patch-clamp recordings (Yeung et al., 2025), sampled with replacement within type from a donor-disjoint cohort. Development used 298 fitted cells and confirmation used a disjoint set of 299 cells; no donor contributes to both. Each cell has $K = 8$ input and spike-history filters (time constants ≈0–2 s) and a fitted static nonlinearity, simulated at the fitted $\Delta t = 20$ ms:

$$h_{i,t}^{(k)} = (1 - d_k)h_{i,t-1}^{(k)} + a_{ik}I_{i,t} + \beta_{ik}r_{i,t-1}, \qquad r_{i,t} = \gamma_i \,\mathrm{ReLU}\big[\tanh\big(\textstyle\sum_d c_{id}^2 x_{i,t}^d\big)\big], \tag{1}$$

with $x_{i,t} = (\bar{h}_{i,t} - b_i)/\sigma_i$. Intrinsic parameters are never trained. A label-free current sweep sets each cell's isolated baseline to 2% of its fitted ceiling. The input current is

$$I_{i,t} = I_i^0 + S_i\Big(\textstyle\sum_\ell W_{i\ell}^{\mathrm{LGN}}(g_{\ell,t} - \bar{g}_\ell^{\mathrm{gray}}) + \sum_j s_j|W_{ij}|\, r_{j,t-1}/R_0 + \xi_{i,t}\Big), \tag{2}$$

with Dale signs s_j, $R_0 = 100$ Hz and Gaussian background noise ξ ($\sigma = 0.15$).

Connectivity. Peak connection probabilities between L2/3 excitatory, PV, SST and VIP cells were taken from the Allen V1DD electron-microscopy estimates used by Ito et al. (2026) (their `pmax_matrix_v1dd.csv`). Excitatory-to-excitatory probability was modulated by orientation difference with their like-to-like rule ($\beta = 0.5$). Magnitudes are log-normal ($\sigma_{\log} = 0.8$) and scaled by the median 90th-percentile PSP of each pre/post type pair from the Allen synaptic-physiology survey (Campagnola et al., 2022). They are normalized by in-degree, with inhibitory sources ×2.

Visual input. Stimuli are rendered on a 48×64 grid (2°/pixel). The LGN front end has ON and OFF centre-surround units (difference of Gaussians, 2°/6°) on a stride-2 grid, each with a sustained (40 ms low-pass) and a transient (20−80 ms band-pass) channel, rectified around a baseline (3,072 units). LGN→V1 weights are fixed. Each Pyr cell pools ON and OFF subfields of a Gabor template at its preferred orientation and receptive-field position. Each subfield draws on sustained or transient channels at random, which yields direction preference (Lien and Scanziani, 2018). Interneurons receive broader, weaker and mostly untuned input.

4.2 Objectives

Data matching (following Ito et al., 2026): one batch has four gray-screen trials and two trials per direction of an 8-direction drifting grating (0.04 cpd, 2 Hz, 80% contrast), each 500 ms after 200 ms of warm-up. For each cell type, the loss is the squared 1-D Wasserstein distance between model and in-vivo distributions of log spontaneous rate, log preferred-direction rate, OSI and DSI. OSI and DSI use the form of Ito et al. (2026) and are computed only for cells responding at $\geq$0.5 Hz. In-vivo targets are the curated Visual Coding Neuropixels units of Ito et al. (2026) (Siegle et al., 2021): L2/3 excitatory (503 units) and L2/3 PV (133) cells, and SST (70) and VIP (16) cells pooled over layers.

Change detection follows the structure of the Allen Visual Behavior task: gray, image A for 240 ms, 500 ms gray, image B for 240 ms. The network reports whether B differs from A from the mean readout of Pyr cells 100–500 ms after B onset. Images come from a fixed set of eight natural movie frames, chosen to be maximally distinct, as mice in that task see eight images. A weak homeostatic term keeps each cell's mean rate near its calibrated baseline. No image used for evaluation is ever used for training.

4.3 Learning rules

Only the recurrent magnitudes $|W_{ij}|$ (fixed mask and sign) and the readout learn. All rules use Exponentiated Adam (Ito et al., 2026; Cornford et al., 2024): $|W| \leftarrow |W| \exp(-\eta\, \hat{m}/(\sqrt{\hat{v}} + \epsilon))$. This update is multiplicative and sign-preserving.

BPTT uses the exact gradient through every filter, spike-history loop and recurrent path.

Local (e-prop; Bellec et al., 2020) uses $\hat{g}_{ij} = \sum_t L_{i,t}\, \partial r_{i,t}/\partial |W_{ij}|$, where the derivative follows only cell i's own filters and spike-history loop and treats other cells' activity as given input. We compute it with automatic differentiation by detaching presynaptic activity in the synaptic input path. Forward values are unchanged; with no recurrent paths the estimate equals BPTT exactly (unit test). The learning signal $L_{i,t}$ is the cell's direct error for the data objective. For change detection it is the readout error broadcast through a fixed random feedback vector.

Local (no β) additionally detaches the spike-history path.

Readout keeps the recurrent weights fixed.

Untrained is the same circuit before learning.

Every rule used the same learning rates, chosen on the development cohort by validation loss: $\eta_{\text{rec}} = 0.015$ for data matching (best for every rule in $\{0.002, 0.005, 0.015\}$; BPTT was already unstable late in training at this value, so larger values were not tried) and $\eta_{\text{rec}} = 0.005$, $\eta_{\text{out}} = 0.3$ for change detection (the readout rate was best for every rule in $\{0.1, 0.3, 1\}$; the recurrent rate was not tuned). Training ran for 1,500 (data) or 800 (change) updates, and the checkpoint with the best validation value was evaluated.

4.4 Evaluation and comparison with mouse V1

After training, every network saw: - the 118 Brain Observatory natural scenes, 240 ms each, back to back in a new random order on each of 10 repeats, as in the recordings (Siegle et al., 2021; de Vries et al., 2020); - 2-s drifting gratings in 8 directions; - 2-s gray screen.

Mouse data. We used VISp units passing Allen quality criteria (ISI violations < 0.5, amplitude cut-off < 0.1, presence ratio > 0.9) from the Visual Coding Neuropixels sessions with the most V1 units. For each unit we took spike counts 0–250 ms after each of the 50 presentations of each scene. Units were pooled into a pseudo-population; regular- and fast-spiking units were separated at a 0.4 ms trough-to-peak waveform duration.

Geometry. - *RDM*: 1 minus the Pearson correlation between z-scored population vectors (Kriegeskorte et al., 2008). - *RSA*: Spearman correlation between the upper triangles of the model and mouse RDMs. We also report a disattenuated version that divides by the square root of the product of the two RDMs' Spearman–Brown-corrected split-half (repeat) reliabilities. - *CKA*: linear CKA (Kornblith et al., 2019) on z-scored image×unit matrices. - *Dimensionality*: participation ratio of the image-response covariance with 200 units subsampled. - *Lifetime sparseness* (Vinje and Gallant, 2000). - *Image decoding*: nearest-centroid, leave-one-repeat-out, 200 units and 10 repeats for both model and mouse. - *Tuning distributions*: similarity to the in-vivo distributions per cell type (1 minus the Kolmogorov–Smirnov statistic). - *Geometry along training*: to separate learning-rule effects from task performance, we recomputed RSA and CKA at every saved checkpoint and related them to that checkpoint's validation value.

Statistics. The unit of analysis is the network seed (10 per condition, paired across rules). We used exact two-sided sign-flip tests and percentile bootstrap CIs, with Holm correction within each objective's primary family. The protocol was locked before the confirmation runs (`docs/PREREGISTRATION_v12.m` `docs/LOCK_v12.json`).

Code and data availability

All code, configurations, the locked protocol (`docs/PREREGISTRATION_v12.md`, `docs/LOCK_v12.json`) and per-network results are in the project repository. The fitted neuron models come from Yeung et al. (2025). Connectivity tables and curated Neuropixels statistics come from Ito et al. (2026). Neuropixels recordings and natural-scene images are from the Allen Brain Observatory (Siegle et al., 2021).

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